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Modelling and simulation of TSV considering void and leakage defects

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Abstract
Through-silicon-via (TSV) technology represents a significant advancement in the fabrication of three-dimensional (3D) integrated circuits, enabling the vertical interconnection of chips. This process results in several defects that impact the signal transmission performance of TSVs. This study establishes a unified equivalent circuit model that includes leakage defect TSV, void defect TSV, and defect-free TSV, using a distributed modelling approach. The established equivalent circuit model is then simulated, and its accuracy is confirmed by comparing the S-parameter values with those from 3D electromagnetic simulator simulations. The impact of defects on the transmission performance of TSV signals was investigated by varying the dimension of the leakage factor, the position of the leakage defects, and the voiding factor and void defect position. Additionally, the impact of the coexistence of void and leakage defects on TSV signal transmission performance is investigated.

Keywords Through-silicon via (TSV) · Void defect · Leakage defect · Equivalent circuit model

1 Introduction

Three-dimensional integrated circuits based on TSV connect multi-layer chips vertically through the TSV structure, significantly reducing the length of the interconnect line and improving chip integration. The benefits of TSV technology in terms of higher system reliability, reduced latency and low power consumption have made it one of the essential technologies for the development of 3D integrated circuits [1–6].

Given the inherent complexity of the manufacturing process, it is inevitable that TSVs will exhibit numerous defects. Figure 1 displays scanning electron microscope (SEM) images of leakage and void defects [7, 8]. The high aspect ratio of the TSVs presents a challenge in the copper electroplating process, as it complicates the deposition of the metal. The considerable aspect ratio of TSVs results in incomplete filling of the copper vias, which in turn gives rise to void defects. These defects partially disconnect the conductive path, thereby creating incomplete open-circuit defects. The presence of contaminants or uneven dielectric deposition within the insulating layer can result in the formation of leakage defects. Furthermore, an insufficient thickness of the insulating layer, susceptible to mechanical and thermal stress, can result in the emergence of leakage defects [9–19].

In [20], a comprehensive equivalent circuit model of defect-free TSV is created after taking the skin effect into account. In [21], a formula for determining the equivalent resistance of TSV for void defect is proposed. The resistance and capacitance values of TSVs with 20 different void radii are extracted. The simulation values are compared with the proposed calculation formula in order to verify the equivalent circuit model of void defect TSV. In [22], a void defect equivalent circuit model of TSV that incorporates the effect of the silicon substrate is proposed. Yuling Shang develops an equivalent circuit model for void defect and leakage defect, which takes into account the silicon substrate effect, the proximity effect, and the skin effect of conductors at high frequencies [23]. Nevertheless, these equivalent circuit models are inadequate for accurately describing circuit models in the presence of numerous defects, as they are lumped modelling approaches. Yu-Jung Huang develops TSV models of single-chip and multi-chip stacked structures [24]. The study employs a systematic approach to investigate the impact of varying proportions of void defect on the S-parameters of the TSV structure. The modelling and simulation of the GSG (Ground-Signal-Ground) TSV-RDL (ReDistribution Layer) are conducted by LUO [25]. The defects are categorised as either randomly distributed in the TSV or at TSV-RDL or

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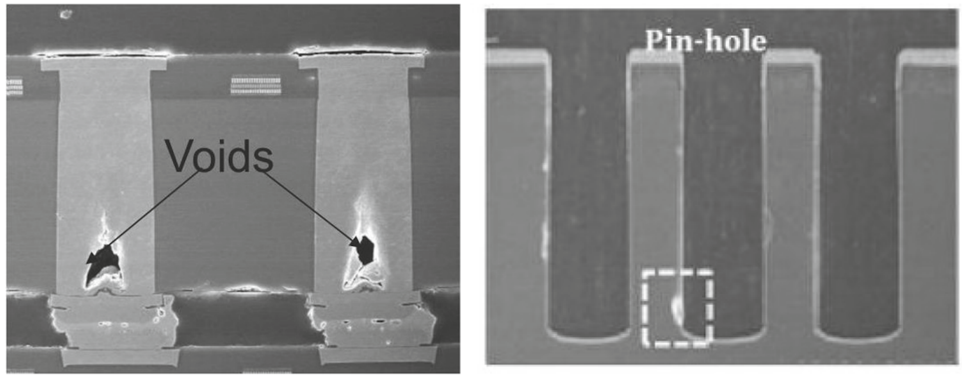


(a) (b)

Fig. 1 SEM images of void defect and leakage defect

TSV-microbump junctions. In [26], the shift in the S-parameters of the GSG-TSV structure at 40 GHz is investigated. In the modelling process, the influence of coplanar waveguides is taken into account. Nevertheless, the simulation of void and leakage defects is based on a 3D electromagnetic simulator, and there is no simulation of an equivalent circuit model for comparison and verification. Furthermore, the modelling and simulation of a single defect are conducted, yet the situation where multiple defects exist simultaneously is not addressed.

This paper examines the influence of void and leakage defects within TSVs on the transmission characteristics of TSVs. A distributed modelling approach is employed to establish a unified equivalent circuit model that encompasses defect-free TSV, void defect TSV, and leakage defect TSV. The accuracy of the equivalent circuit model is confirmed by comparing the results of the simulation with those of the three-dimensional electromagnetic simulator. Three-dimensional electromagnetic simulations are conducted using the HFSS software. HFSS boasts powerful electromagnetic field solving capabilities and precise design verification, enabling the simulation of intricate geometric structures and material properties. It is capable of accurately predicting the electromagnetic performance of devices, effectively reducing experimental costs and design cycles, optimising performance, and ensuring the feasibility and efficiency of the design. The top and bottom extremities of the defective TSV are designated as wave ports. Radiation boundary conditions are applied to the model, and the mesh is refined using adaptive mesh refinement technique of HFSS. The S-parameters, specifically the quantities S_{11} and S_{21}, are employed to characterise the reflection and transmission phenomena occurring within a network. The parameter S_{11} quantifies the reflection of a signal at port 1, thereby indicating the degree of impedance matching. In contrast, S_{21} provides a measure of the transmission of a signal from port 1 to port 2. A high value of S_{11} indicates poor matching, whereas a high value of S_{21} suggests efficient transmission. These measurements provide insights into the performance of the TSV. Concurrently, an increase in void factor will result in a deterioration of the transmission characteristics of TSV, and a change in void location has a negligible impact on the transmission characteristics of TSV. The increase in the leakage factor and the position of the leakage defect in close proximity to the source end will result in a slight deterioration in the signal transmission performance of the TSV. In the event that both void defects and leakage defects are present within the TSV, it can be observed that the leakage defect exerts a predominant influence on the TSV transmission performance.

The remainder of this paper is organised as follows: In Sect. 2, a distributed equivalent circuit model of the defect TSV is constructed. In Sect. 3, an equivalent circuit model is employed to investigate the impact of single defect dimension and position on TSV transmission performance. Furthermore, the impact of the coexistence of void and leakage defects on the S-parameters of TSV is also considered. Finally, the conclusions are presented in Sect. 4.

2 Construction of the equivalent circuit model of defective TSV

The most prevalent configuration is a cylindrical TSV filled with copper metal and covered with a thin silica insulation layer. This serves to reduce crosstalk and prevent leakage. Figure 2a shows the cylindrical TSV, as well as the void and leakage defects structures. A distributed RLGC parametric equivalent circuit model can be used to represent TSVs. In high-frequency applications, the inductance, capacitance, and transmission line effects of the circuit become significant. The lumped parameter model assumes that these effects can be described at a few points, whereas the distributed model considers the continuous distribution of these effects throughout the circuit. Consequently, distributed models are more effective in accurately portraying the behaviour of high-frequency circuits. Concurrently, the distributed model is more effective in simulating the propagation and reflection

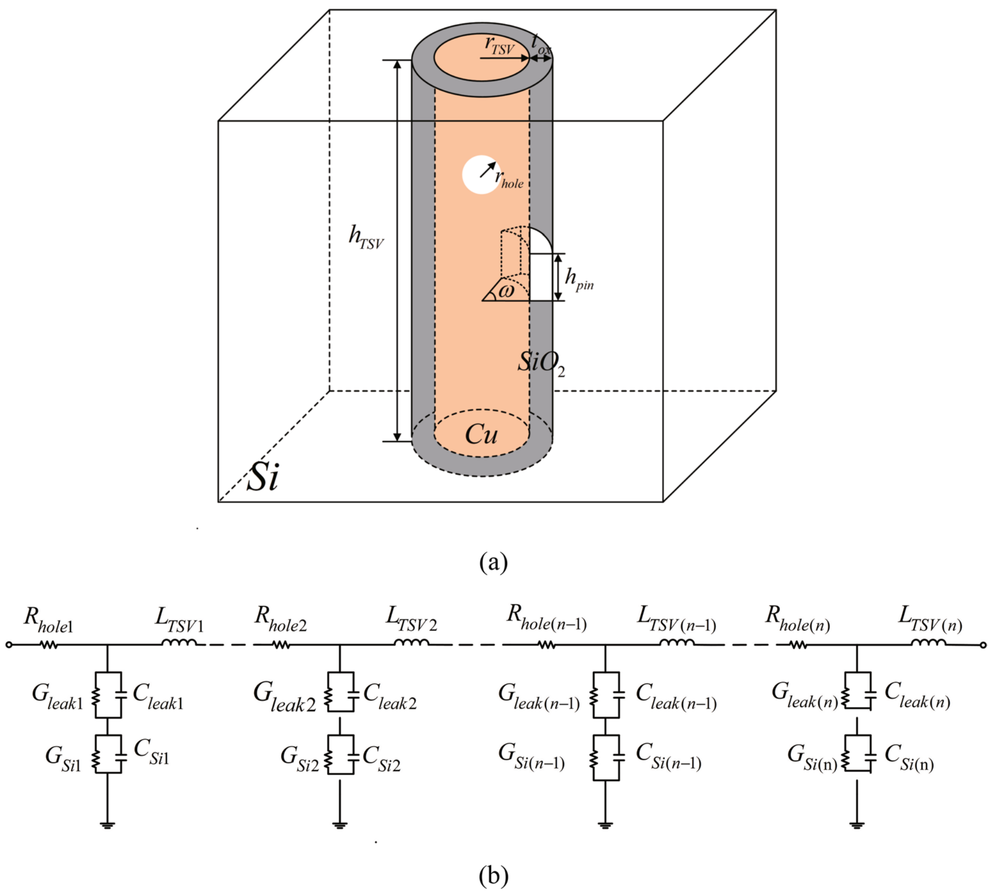


Fig. 2 **a** Defect TSV structure diagram, **b** defect TSV equivalent circuit diagram

of signals within the actual circuit, due to its incorporation of distributed parameters.

The distributed model divides the TSV into n aliquots, employs a lumped model to represent each part, and connects them in series to characterise the characteristics of the entire TSV, as shown in Fig. 2b. The parameters of the model are the resistance R, inductance L, leakage conductance G, and capacitance C of the TSV. For a cylindrical copper TSV with radius r_{TSV}, height h_{TSV}, and resistivity ρ_{Cu}, the height of each part of the TSV is l_{TSV}, where $h_{TSV} = n \times l_{TSV}$. As shown in Fig. 2a, the void defect is modelled as fabricating a spherical air void in the centre of the defect-free TSV with a void radius r_{hole}. The leakage defect is modelled by fabricating an arc air void on the insulation layer of a defect-free TSV, where ω is the arc of the arc and h_{pin} is the height of the arc [26].

The variable l_{TSV} represents the height of each part of the TSV, where $l_{TSV} = \max\{2r_{hole}, h_{pin}\}$. In the process of segmentation, the focus is directed towards the minor segment that encompasses the entirety of the defect, and this segment is then divided both upwards and downwards. The variables n_1 and n_2 represent the number of segments on the upper and lower sides, respectively, where $n_1 = x(h_{TSV}/l_{TSV}) - 0.5$ and $n_2 = (1-x)(h_{TSV}/l_{TSV}) - 0.5$. The total number of segments $n = n_1 + n_2 + 1$, where n_1 and n_2 are integers. The total number of segments $n = [n_1] + [n_2] + 1 + x_0$, where n_1 or n_2 are not integers, $[n]$ is the integer function, x_0 is the number of n_1 and n_2 are not integers. Let x be the normalisation parameter for the height of TSV. If x=*0*, this indicates that the defect is located at the upper extremity of the TSV. If x=*0.5*, this signifies that the defect is situated in the middle of the TSV. If x=*1*, this denotes that the defect is positioned at the lower extremity of the TSV.

In accordance with the established formula for calculating the DC resistance of a cylindrical conductor, the DC resistance of TSV can be expressed as follows [12]:

$$R_{TSV-dc} = \rho_{Cu} \frac{l_{TSV}}{\pi r_{TSV}^2} \tag{1}$$

As the frequency of operation increases, the current in the conductor undergoes a more rapid change, resulting in an uneven distribution of current density across the cross-section of the TSV. The current is concentrated near the surface of the conductor, leading to the formation of a skin effect, which increases the resistance of the conductor. In light of the influence of the skin effect, the formula for calculating the high-frequency parasitic resistance of TSV is as follows [12]:

$$R_{TSV-ac} = \rho_{Cu} \frac{l_{TSV}}{\pi\delta(2r_{TSV} - \delta)} \tag{2}$$

$$\delta = \sqrt{\frac{\rho_{Cu}}{\pi f \mu_0 \mu_{r,Cu}}} \tag{3}$$

where δ is the skin depth of the TSV, f is the operating frequency of the chip, and μ_0 is the dielectric constant in the vacuum. Therefore, the total resistance of the TSV is [12]:

$$R_{TSV} = \sqrt{R_{TSV-dc}^2 + R_{TSV-ac}^2} \tag{4}$$

The void factor, represented by parameter m, is defined as the ratio of the void radius to the TSV radius, where $m = r_{hole}/r_{TSV}$ and void factor m is employed to quantify the size of the void defect. The void defect will affect the TSV resistance of the specified segment, while the TSV resistance of other segments remains unaltered. Assuming the void defect is located at a height z with a cross-sectional area of $S = \pi(r_{TSV}^2 - r_{hole}^2 + z^2)$, the total void resistance can be obtained by integrating the height of the void defect. The expression is as follows [23]:

$$R_{hole-dc} = \rho_{Cu} \frac{l_{TSV} - 2mr_{TSV}}{\pi r_{TSV}^2} + \frac{2\rho_{Cu}}{\pi\sqrt{r_{TSV}^2(1-m^2)}} \arctan\left(\frac{mr_{TSV}}{\sqrt{r_{TSV}^2(1-m^2)}}\right) \tag{5}$$

$$R_{hole-ac} = \rho_{Cu}\left[\frac{l_{TSV} - 2mr_{TSV}}{\pi\delta(2r_{TSV} - \delta)} + \frac{2mr_{TSV}}{\pi r_{TSV}^2(1-m^2)}\right] \tag{6}$$

$$R_{hole} = \sqrt{R_{hole-dc}^2 + R_{hole-ac}^2} \tag{7}$$

As demonstrated by the expressions of equations (5), (6), and (7), the absence of void defect within the segmented TSV, characterised by m=0, results in a consistency between the aforementioned equations and those expressed by equations (1), (2), and (4). Consequently, equations (5), (6), and (7)can be used to represent both defect-free TSV and void defect TSV. The change in TSV resistance with the voiding factor m is illustrated in Fig. 3.

As shown in Fig. 3, when the radius of the void is relatively small, any changes in the void resistance are insignificant. In this instance, the skin effect predominantly governs the current flow along the surface of the TSV. Here, the AC resistance

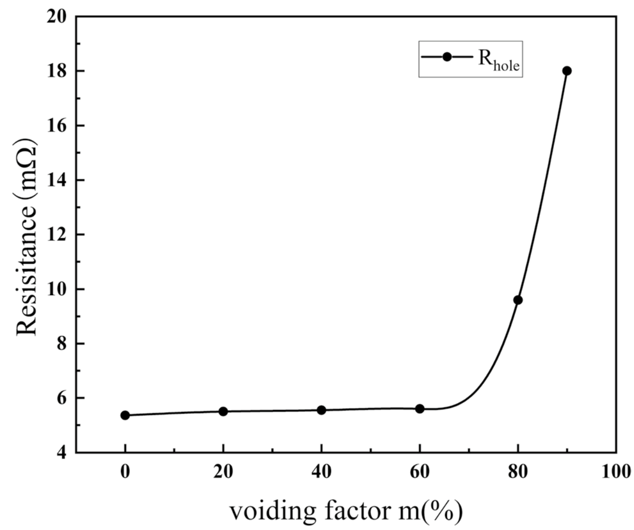


Fig. 3 Variation curves of the resistance of TSV R_{hole} with the void factor m

of the void is minimal, while the DC resistance accounts for the majority. However, as the void factor increases, the AC resistance also rises, eventually becoming the dominant contributor to the void resistance. As the void factor increases, that is to say, as the void radius approaches the TSV radius, the TSV resistance undergoes a significant change, exhibiting exponential growth.

The inductance of the TSV is contingent upon the position of the current loop. In the context of a single TSV analysis, the TSV inductance is calculated on the assumption that the current loop is at infinity. In this instance, the inductance expression is [20]:

$$L_{TSV} = \frac{\mu_0\mu_{r,Cu}}{2\pi}\left[l_{TSV} \ln\left(\frac{l_{TSV} + \sqrt{r_{TSV}^2 + l_{TSV}^2}}{r_{TSV}}\right)\right] + \frac{\mu_0\mu_{r,Cu}}{2\pi}\left(r_{TSV} - \sqrt{r_{TSV}^2 + l_{TSV}^2}\right) \tag{8}$$

where μ_0 is the vacuum permeability. The TSV structure diagram indicates that the TSV, silica insulation layer, and silicon substrate will form a MOS parasitic capacitor. The TSV capacitor is essentially composed of insulation layer capacitance C_{ox}, parasitic MOS capacitance C_{dep}, and substrate capacitance C_{sub}. The calculation formula is as follows [21, 23]:

$$C_{ox} = \frac{2\pi\varepsilon_{r,ox}\varepsilon_0 l_{TSV}}{\ln\left(\frac{r_{TSV}+t_{ox}}{r_{TSV}}\right)} \tag{9}$$

$$C_{dep} = \frac{2\pi\varepsilon_0\varepsilon_{r,Si} l_{TSV}}{\ln\left(\frac{r_{TSV}+t_{ox}+w_{dep}}{r_{TSV}+t_{ox}}\right)} \tag{10}$$

$$w_{dep} = \sqrt{\frac{4\varepsilon_0\varepsilon_{r,Si}V_{th}\ln(N_A/n_i)}{qN_A}} \tag{11}$$

$$C_{sub} = \frac{2\pi\varepsilon_{r,si}\varepsilon_0 l_{TSV}}{\ln(\frac{r_{substrate}}{r_{TSV}+t_{ox}})} \tag{12}$$

$$C_{Si} = \frac{C_{sub}C_{dep}}{C_{sub}+C_{dep}} \tag{13}$$

where, ε_0 is the vacuum dielectric constant, $\varepsilon_{r,ox}$ is the dielectric constant of the silica insulating layer, t_{ox} is the thickness of the insulating layer, w_{dep} is the width of the depletion layer on the semiconductor side, N_A is the impurity concentration, n_i is the intrinsic carrier concentration, V_{th} is the thermal voltage, q is the amount of electronic charge, C_{Si} is the total parasitic capacitance of the substrate, and $r_{substrate}$is the length of the substrate.

TSV conductance mainly refers to the leakage conductance of the insulation layer and the parasitic conductance of the silicon substrate. The magnitude of the leakage conductance of the insulating layer depends on the resistivity of the insulating material ρ_{SiO_2}, the thickness of the insulating layer, and the magnitude of the substrate conductance depends on the conductivity of the substrate σ_{Si}. The calculation formula is as follows [21]:

$$G_{INS} = \frac{2\pi\sigma_{SiO_2}l_{TSV}}{\ln(\frac{r_{TSV}+t_{ox}}{r_{TSV}})} \tag{14}$$

$$G_{Si} = \frac{2\pi\sigma_{Si}l_{TSV}}{\ln(\frac{r_{substrate}/2}{r_{TSV}+t_{ox}})} \tag{15}$$

The parameter j is the leakage factor, defined as ratio of the area of the leakage defect to the area of the TSV insulation layer, where $j = \omega h_{pin}/2\pi r_{TSV}$ and leakage factor j is employed to quantify the size of the leakage defect. In theory, the impedance between the TSV and the silicon substrate is infinite. However, in the event of a leakage defect occurring in the insulation, a leakage current may be generated between the TSV and the substrate, resulting in a reduction in the breakdown voltage and an increase in leakage conductance. Leakage conductance G_{leak} and insulating layer capacitance C_{leak} are as follows [23]:

$$G_{leak} = \frac{2\pi\sigma_{SiO_2}l_{TSV} - 4\pi^2 jf\varepsilon_0\varepsilon_{r,air}r_{TSV}}{\ln(\frac{r_{TSV}+t_{ox}}{r_{TSV}})} \tag{16}$$

$$C_{leak} = \frac{2\pi\varepsilon_{r,ox}\varepsilon_0(l_{TSV} - jr_{TSV})}{\ln(\frac{r_{TSV}+t_{ox}}{r_{TSV}})} \tag{17}$$

where $\varepsilon_{r.air}$ is the relative permittivity of air. When j=0, the aforementioned expressions for leakage conductance and insulation capacitance are consistent with those of defect-free TSV. Consequently, formulae (16) and (17) may be employed to characterise both defect-free and leakage defect TSVs with respect to leakage conductance and insulation capacitance. As shown in Fig. 4, the leakage conductance and insulation capacitance are contingent upon the leakage factor j.

As can be seen from Fig. 4, in the case of leakage defects, even if the leakage factor is only 5%, the leakage conductance of the TSV increases significantly from 3×10^{-15} to $8.5\mu S$. With the increase of the leakage factor j, that is, the increase of the leakage area, the leakage conductance increases nearly linearly. The capacitance of the insulation layer decreases linearly with the increase of the leakage factor. The increase of leakage conductance is caused by the increase of the leakage factor, which makes the insulation between the TSV and the substrate worse, and the decrease of the capacitance of the insulation layer is caused by the decrease of the insulation area due to the increase of the leakage factor.

Based on the above analysis, when $m = 0, j = 0$, defect-free TSV can be characterised. When $m \neq 0, j = 0$, the void defect TSV can be characterised. When $m = 0, j \neq 0$, the leakage defect TSV can be characterised. When $m \neq 0, j \neq 0$, it can be characterised that both void defects and leakage defects exist.

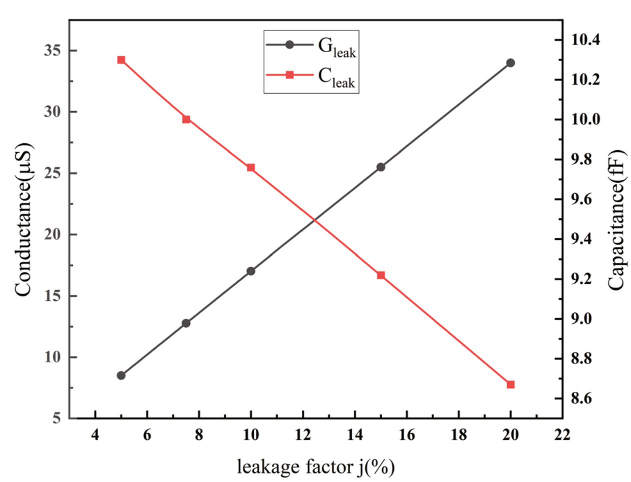


Fig. 4 Variation curves of leakage conductance G_{leak} and insulation layer capacitance C_{leak} with leakage factor j

3 Simulation and validation

In this section, the equivalent circuit model is subjected to simulation and compared with the results of the 3D electromagnetic simulator. The HFSS software enables the precise modelling of defective TSVs, thereby guaranteeing uniformity between the three-dimensional structural model and the equivalent circuit model of the defective TSV. In order to ascertain the dimensions and location of the defect with precision, the impact of modifying the dimensions and location of the defect on the transmission characteristics of TSV was investigated. Furthermore, the impact of the coexistence of void and leakage defects on the S-parameters of TSVs is analysed.

3.1 Simulation of defect-free TSV

Table 1 presents the structural parameters of the complete defect-free TSV model, as reported in the literature [27]. The TSV parameters of the model are consistent.

Table 1 Parameters of the ideal defect-free TSV model

Parameter	Definition	Value
r_{TSV}	radius of TSV	2.47 μm
h_{TSV}	height of TSV	25 μm
t_{ox}	thickness of the insulating layer	0.13 μm
$r_{substrate}$	length of the substrate	40 μm
σ_{Si}	conductivity of the substrate	$10\, S/m$
ρ_{Cu}	resistivity of Cu	$1.68 \times 10^{-8}\,\Omega \cdot m$
$\varepsilon_{r,Si}$	relative permittivity of silicon	11.9
$\varepsilon_{r,ox}$	relative permittivity of insulator	4
$\mu_{r,Cu}$	relative permeability of Cu	1
δ_{ox}	the loss tangent of insulating layer	0
δ_{Si}	the loss tangent of Silicon	0

In order to study the influence of defects on the signal transmission performance of TSV, the simulation results of defect-free TSV are used as a reference. The equivalent circuit model of the defect-free TSV is compared with the results of the 3D electromagnetic simulator in Fig. 5. As can be seen from Fig. 5, at 100 GHz, the S_{11} amplitude obtained by the equivalent circuit model is −16.5 dB, and the S_{11} amplitude obtained by the 3D electromagnetic simulator is −16.2 dB, with an error of 0.3 dB. The S_{21} amplitudes are −0.31 dB and −0.326 dB, respectively, with an error of 0.013 dB. The degree of fitting of the S_{11} and S_{21} curves is above 96%, which indicates the effectiveness of the defect-free TSV equivalent circuit model.

3.2 Simulation of void defect TSV

The void defect can be established by creating a spherical air void in the centre of the defect-free TSV with an initial radius. In order to ascertain the impact of alterations to the physical parameters of the void defect on the electrical transmission performance of the TSV, the control variable method was employed to simulate the void defect. In the simulation, the controllable variables are the voiding factor m, which is the radius of the void defect, and the normalised position x. As illustrated in Table 2, the model parameters for the void defect TSV are outlined below.

The 3D simulation FEM mesh and current density of the void defect TSV are illustrated in Fig. 6. It is evident that the FEM mesh exhibits increased density at the void defect. The current density of the entire void defect TSV is approximately uniformly distributed. The frequency variation curves of the S-parameters of TSV under different voiding factors are shown in Fig. 7.

As shown in Fig. 7, in the low-frequency range, the S_{11} and S_{21} curves generated by the equivalent circuit model exhibit a high degree of overlap with the curves generated

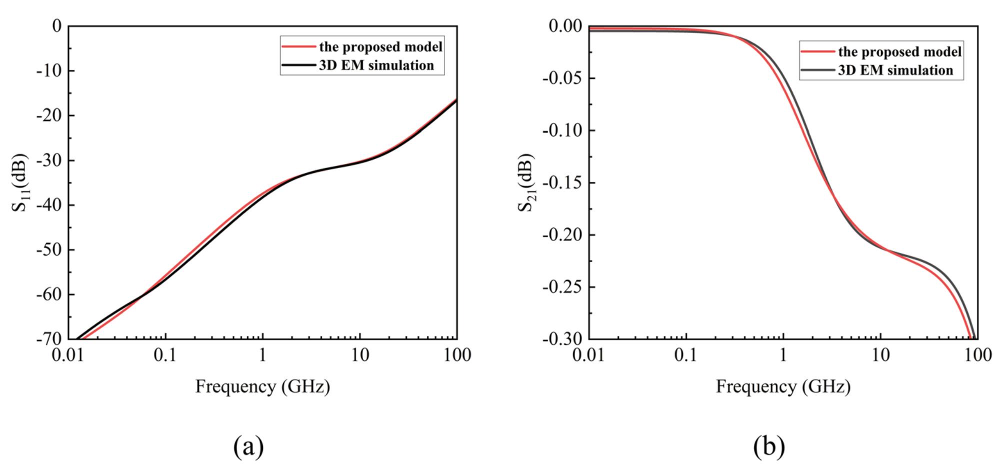


Fig. 5 S-parameter plots of defect-free TSV **a** S_{11} curve and **b** S_{21} curve

Table 2 Parameters of the void defect TSV model

	$r_{hole}/\mu m$	$x_{hole}/\mu m$
$m = 20\%$	0.494	12.5
$m = 40\%$	0.988	12.5
$m = 60\%$	1.482	12.5
$m = 80\%$	1.976	12.5
$m = 90\%$	2.223	12.5
$x = 0.1$	0.988	22.5
$x = 0.2$	0.988	20
$x = 0.3$	0.988	17.5
$x = 0.4$	0.988	15
$x = 0.5$	0.988	12.5

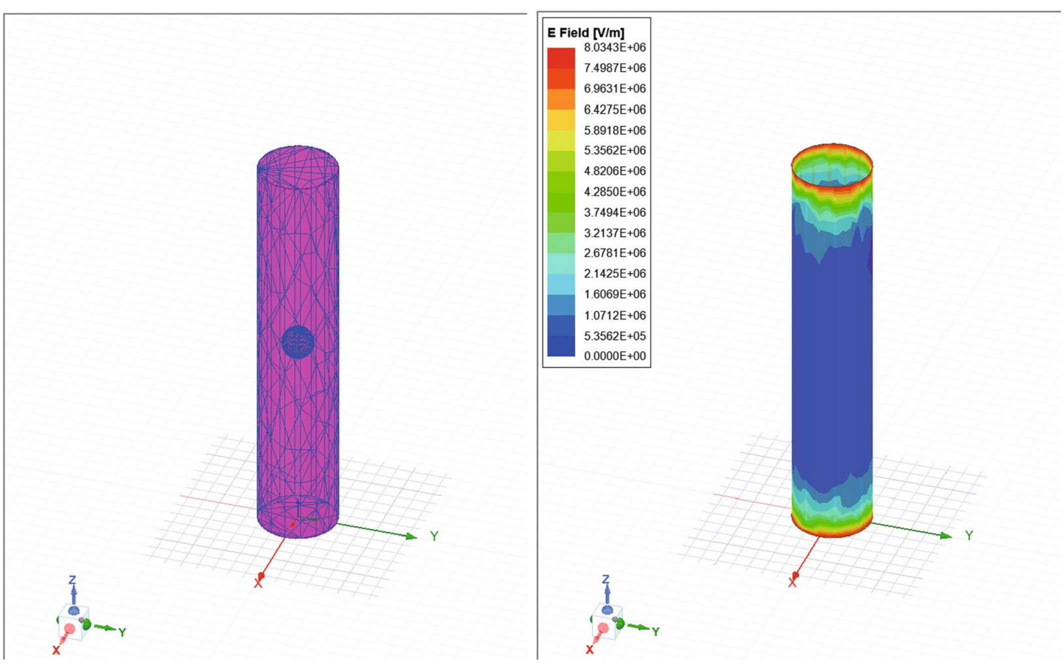
by the 3D electromagnetic simulator. In the high-frequency range, the S_{11} curve derived from the equivalent circuit model exhibits a slight decrease in comparison with the S_{11} curve generated by the 3D electromagnetic simulator, while the S_{21} curve displays a slight increase. The S-parameter curves obtained by the two methods exhibit satisfactory fitting results, thereby corroborating the validity of the equivalent circuit model proposed in this study.

Concurrently, it can be observed that as the voiding factor increases, the S_{11} curve and the S_{21} curve exhibit a high degree of overlap. When the void factor is minimal, that is, the void radius is negligible in comparison with the TSV radius, the S_{11} amplitude and the S_{21} amplitude of the void defective TSV are nearly identical to those of the defec-free TSV. As the voiding factor increases to a point where the voiding defects are almost tangent to the TSV, the S_{11} and S_{21} amplitude undergo slight changes. The S_{11} amplitude experiences an increase, while the S_{21} amplitude undergoes a decrease. A decline in the quality of the TSV signal transmission was observed.

The primary underlying mechanism responsible for this phenomenon is the skin effect, which describes the preferential flow of current primarily through the surface of the conductor. In the event that a void defect exists within the TSV, the signal is still capable of traversing the surface portion of the TSV that has not been truncated. In the presence of void defects that almost completely cut off the TSV, the S parameters of the device in question do not undergo a significant change.

Figure 8 shows the frequency variation curves of the S-parameters of the TSV at different void defect positions. As the position of the void in relation to the source end is altered, the amplitude of S_{11} demonstrates a gradual increase, while the amplitude of S_{21} exhibits a corresponding decrease. This results in a reduction in the quality of the signal transmission performance.

The findings indicate that as the void factor increases and the void location moves closer to the source end, there is a notable increase in the amplitude of S_{11}, an decrease in the amplitude of S_{21}, and a concomitant deterioration in the overall signal transmission performance.



(a) (b)

Fig. 6 3D simulation of void defect TSV: **a** FEM mesh and **b** current density

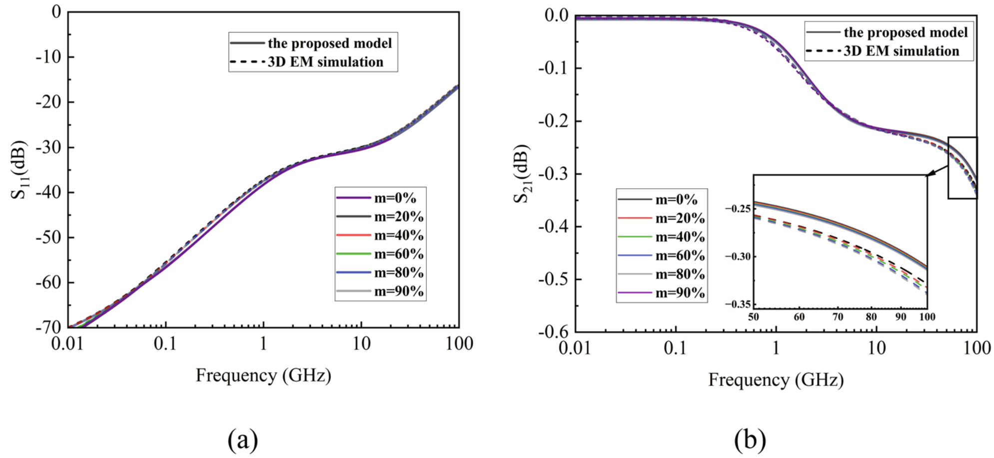


Fig. 7 S-parameter simulation results of TSV for voiding defects under different voiding factors **a** S_{11} curve and **b** S_{21} curve

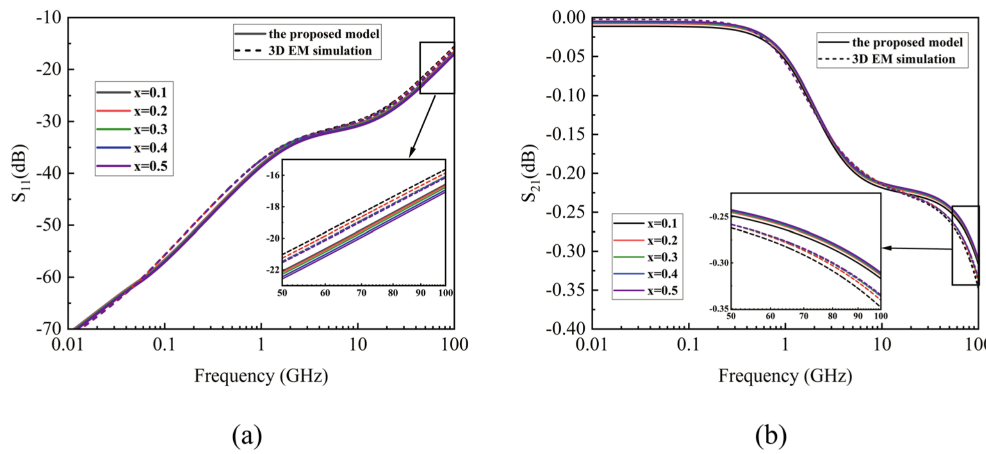


Fig. 8 S-parameter simulation results of TSV of void defects at different position **a** S_{11} curve and **b** S_{21} curve

3.3 Simulation of leakage defect TSV

Leakage defects can be established by fabricating an arc air void on the defect-free TSV insulation, with an initial arc and an initial height. In order to quantify the effect of changes in the physical parameters of the leakage defect on the electrical transmission performance of the TSV, it is necessary to consider the controllable variables. These include the leakage factor j, which represents the area of the leakage defect, and the normalised position x. As illustrated in Table 3, the model parameters for the leakage defect TSV are outlined below.

The 3D simulation FEM mesh and current density of the leakage defect TSV are illustrated in Fig. 9. It is evident that the FEM mesh exhibits increased density at the leakage defect. At the leakage defect, the insulation layer is breached, causing the current to leak into the substrate, resulting in a higher current density. The S-parameters of the TSV with frequency are shown in Fig. 10 under different leakage factors. The S_{11} curve simulated by the equivalent circuit model is in close agreement with the S_{11} curve obtained by the 3D electromagnetic simulator. While the S_{21} curve exhibits minor discrepancies, the overall fitting effect is nevertheless commendable. From the figure, it can be observed that within the low-frequency range, specifically within the 1-GHz frequency range, as the leakage factor j increases, the S_{11} amplitude increases gradually while the S_{21} amplitude decreases gradually in both the equivalent circuit model simulation and the three-dimensional electromagnetic simulation. The signal transmission performance of leakage defect TSV is markedly inferior to that of defect-free TSV.

The primary reason for this phenomenon is the skin effect, which becomes increasingly concentrated on the surface of the TSV as the frequency increases. Leakage defects result in the insulation layer that originally blocked the

Table 3 Parameters of the leakage defect TSV model

	ω	$h_{pin}/\mu m$	$x_{pin}/\mu m$
$j = 5\%$	$\pi/4$	0.988	12.5
$j = 7.5\%$	$\pi/4$	1.482	12.5
$j = 10\%$	$\pi/2$	0.988	12.5
$j = 15\%$	$\pi/2$	1.482	12.5
$j = 20\%$	π	0.988	12.5
$x = 0.1$	$\pi/2$	0.988	22.5
$x = 0.2$	$\pi/2$	0.988	20
$x = 0.3$	$\pi/2$	0.988	17.5
$x = 0.4$	$\pi/2$	0.988	15
$x = 0.5$	$\pi/2$	0.988	12.5

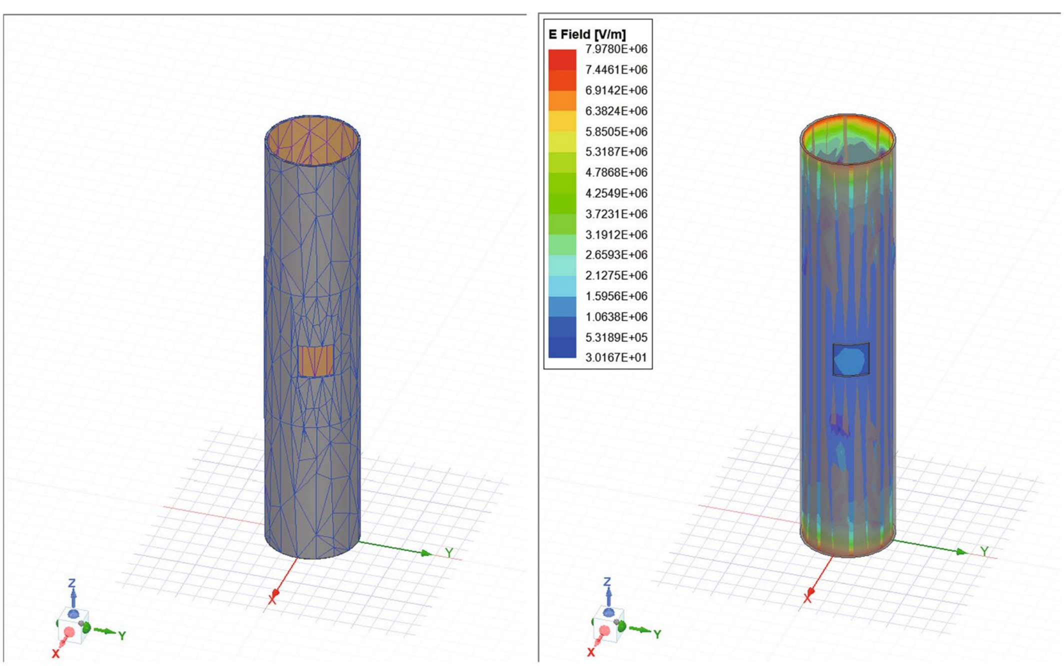
signal being destroyed, allowing the signal to leak into the silicon substrate along the conductive path, thereby causing losses. A larger leakage factor will result in a greater leakage conductance, which will in turn lead to an increased current leakage into the silicon substrate. This will have a detrimental effect on the signal transmission performance of the TSV.

It can be observed that the S_{11} and S_{21} curves exhibit a high degree of overlap under different leakage factors. The reason for this discrepancy is that the insulation between the TSV and the substrate is inadequate due to the leakage defect. However, the leakage resistance is considerably larger than the TSV resistance in milliohms, and the leakage current is minimal.

The S-parameter curve of the TSV with frequency is shown in Fig. 11 at different position of leakage defects. As illustrated in the figure, within the 1-GHz frequency range, the S_{11} amplitude exhibits a gradual increase in value as the location of the leakage defect approaches the source end, while the S_{21} amplitude demonstrates a corresponding gradual decrease. The primary reason is that the further the distance from the source, the smaller the current density, the smaller the current leaking from the leakage defect to the substrate, and the signal transmission performance is marginally enhanced.

The findings indicate that as the leakage factor increases and the leakage location gets closer to the source end, there is a notable increase in the amplitude of S_{11}, accompanied by an decrease in the amplitude of S_{21}. This is accompanied by a deterioration in the overall signal transmission performance. Furthermore, the proximity of the leakage defect to the source end has a significant negative impact on the signal transmission characteristics within the 1-GHz frequency range.

The preceding analysis leads to the following conclusions. The leakage defect will result in a deterioration of the insulation between the TSV and the silicon substrate. Furthermore, an increase in the leakage factor will lead to an increase in the leakage conductance of the TSV and an increase in the current leakage to the substrate. In the low-frequency range, as the leakage factor j increases and the leakage defect location approaches the source end, the S_{11}



(a) (b)

Fig. 9 3D simulation of leakage defect TSV: **a** FEM mesh and **b** current density

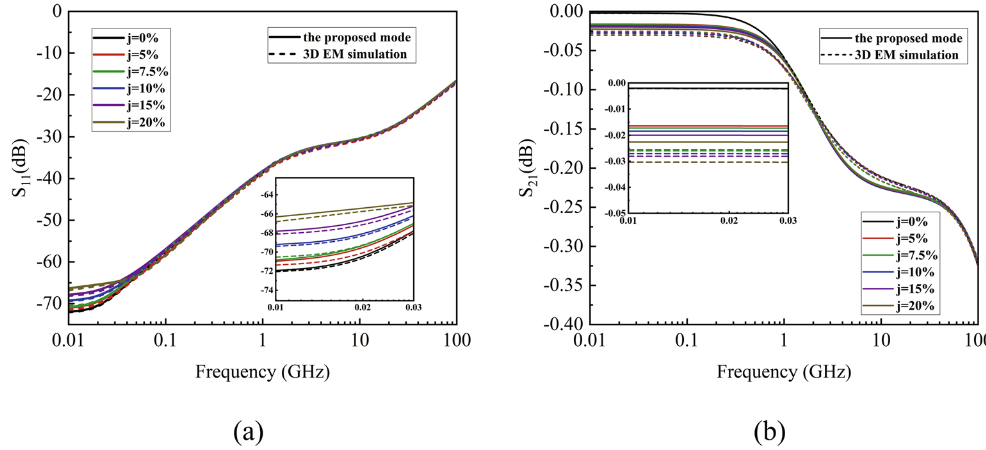


Fig. 10 S-parameter simulation results of TSV for leakage defects under different leakage factors: **a** S_{11} curve and **b** S_{21} curve

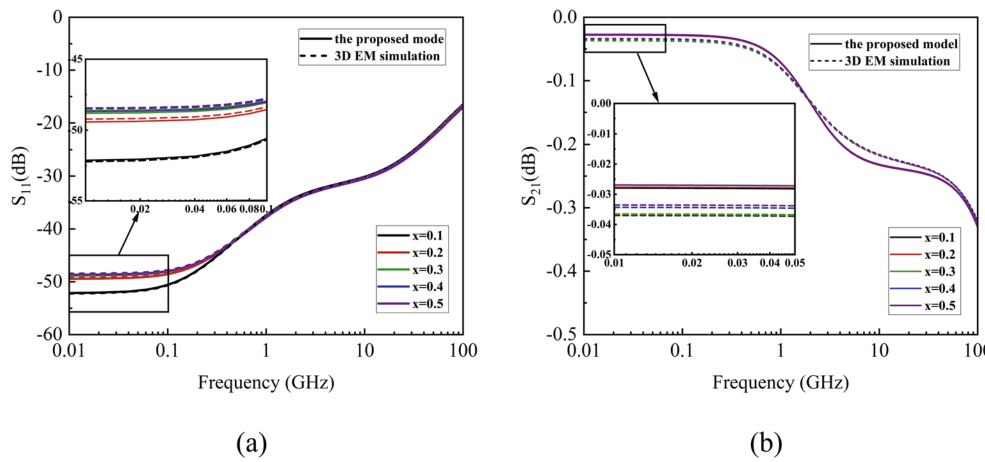


Fig. 11 S-parameter simulation results of TSV for leakage defects at different positions: **a** S_{11} curve and **b** S_{21} curve

amplitude exhibits a gradual increase, the S_{21} amplitude shows a gradual decrease, and the TSV signal transmission performance deteriorates significantly.

3.4 Simulation of the simultaneous existence of void defect and leakage defect

As illustrated in Table 4, the model parameters for the simultaneous existence of void defect and leakage defect TSV are outlined below.

The void defect and the leakage defect are simultaneously at the same section, and the S-parameter simulation results of TSV are presented in Fig. 12. The amplitude of the S_{11} and S_{21} of the TSV remains essentially unaltered when the voiding factor *m* is increased, provided that the leakage factor *j* remains at 5%. When the voiding factor *m* is held constant at 40%, an increase in the leakage factor *j* results in a slight increase in the S_{11} amplitude of the TSV, accompanied by a slight decrease in the S_{21} amplitude in the low-frequency range. This leads to a slight deterioration in signal transmission performance. Consequently, when void defect and leakage defects co-exist in the same section, the signal transmission performance of TSV is primarily influenced by leakage defects.

The dimensions of the voiding factor and leakage factor are altered, respectively, by positioning the void defect at *x=0.3* and the leakage defect at *x=0.5*. The S-parameter simulation results of the TSV are presented in Fig. 13. The S_{11} amplitude and S_{21} amplitude of TSV remain essentially unaltered when the void factor *m* is increased, provided that the leakage factor *j* remains constant at 5%. When the voiding factor *m* is held constant at 40%, an increase in the leakage factor *j* results in a slight increase in the S_{11} amplitude of the TSV, accompanied by a slight decrease in the S_{21} amplitude in the low-frequency range, which in turn leads to a slight deterioration in signal transmission performance. Consequently, when void defects

Table 4 Parameters of the simultaneous existence of void defect and leakage defect TSV model

		$r_{hole}/\mu m$	$x_{hole}/\mu m$	ω	$h_{pin}/\mu m$	$x_{pin}/\mu m$
At the same position		$r_{hole}/\mu m$	$x_{hole}/\mu m$	ω	$h_{pin}/\mu m$	$x_{pin}/\mu m$
	$j = 5\%, m = 0\%$	0	12.5	$\pi/4$	0.988	12.5
	$j = 5\%, m = 20\%$	0.494	12.5	$\pi/4$	0.988	12.5
	$j = 5\%, m = 40\%$	0.988	12.5	$\pi/4$	0.988	12.5
	$j = 10\%, m = 40\%$	0.988	12.5	$\pi/2$	0.988	12.5
	$j = 0\%, m = 40\%$	0.988	12.5	0	0	12.5
At the different position		$r_{hole}/\mu m$	$x_{hole}/\mu m$	ω	$h_{pin}/\mu m$	$x_{pin}/\mu m$
	$j = 5\%, m = 0\%$	0	17.5	$\pi/4$	0.988	12.5
	$j = 5\%, m = 20\%$	0.494	17.5	$\pi/4$	0.988	12.5
	$j = 5\%, m = 40\%$	0.988	17.5	$\pi/4$	0.988	12.5
	$j = 10\%, m = 40\%$	0.988	17.5	$\pi/2$	0.988	12.5
	$j = 0\%, m = 40\%$	0.988	17.5	0	0	12.5

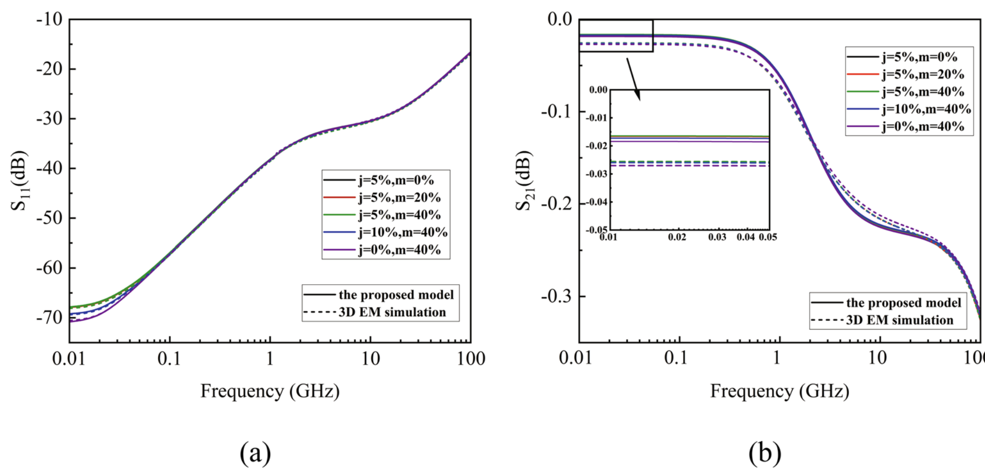


Fig. 12 S-parameter simulation results of TSV with void defect and leakage defect at the same position: **a** S_{11} curve and **b** S_{21} curve

and leakage defects co-exist in the different section, the signal transmission performance of TSV is primarily influenced by leakage defects.

The preceding analysis yielded the following conclusion: in the event of the coexistence of void and leakage defects, the latter will assume the role of the primary factor affecting the performance of TSV transmission.

4 Conclusion

In this paper, a distributed parametric modelling method is employed to construct an equivalent circuit model that integrates defect-free TSV, void defect TSV, and leakage defect TSV. A simulation of the equivalent circuit model was conducted, and the signal transmission performance of the defective TSV at 100 GHz was evaluated by comparing it with the results obtained from the 3D electromagnetic simulator. An increase in the void factor m will result in an elevated void resistance, which will in turn lead to a deterioration in the transmission characteristics of TSV. An increase in the leakage factor j results in a significant enhancement of the leakage conductance, yet the leakage conductance remains considerably smaller than that of the TSV conductance. The signal transmission performance of TSV is observed to deteriorate slightly with the increase in leakage factor j and the leakage defect approaching the source. In the event that both void defects and leakage defects are present simultaneously, the leakage defect will become the primary factor influencing the transmission

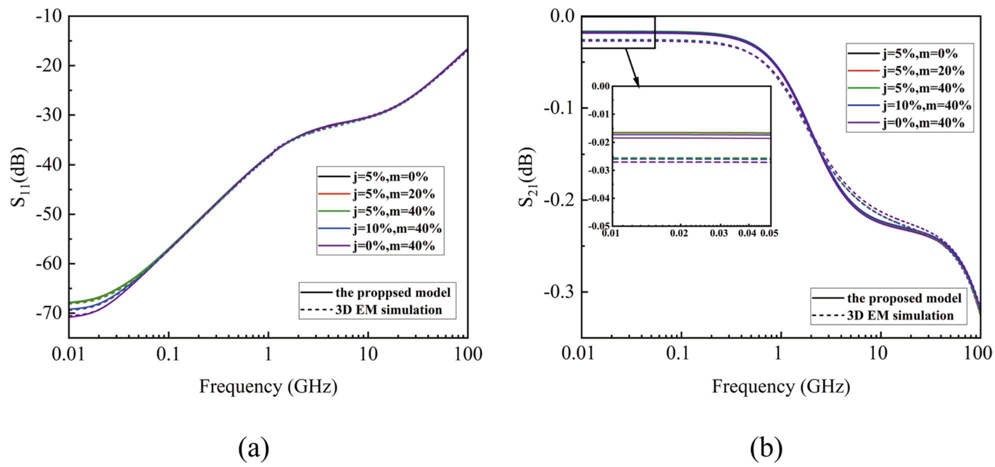


Fig. 13 S-parameter simulation results of TSVs with void defects and leakage defects at different locations: **a** S_{11} curve and **b** S_{21} curve

performance of the TSV. It is frequently observed that the production and utilisation processes result in the formation of defects with irregular geometries. It would therefore be beneficial for future research to focus on exploring the characteristics of TSV defects with a variety of different structures.

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Competing interests The authors declare no competing interests.

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